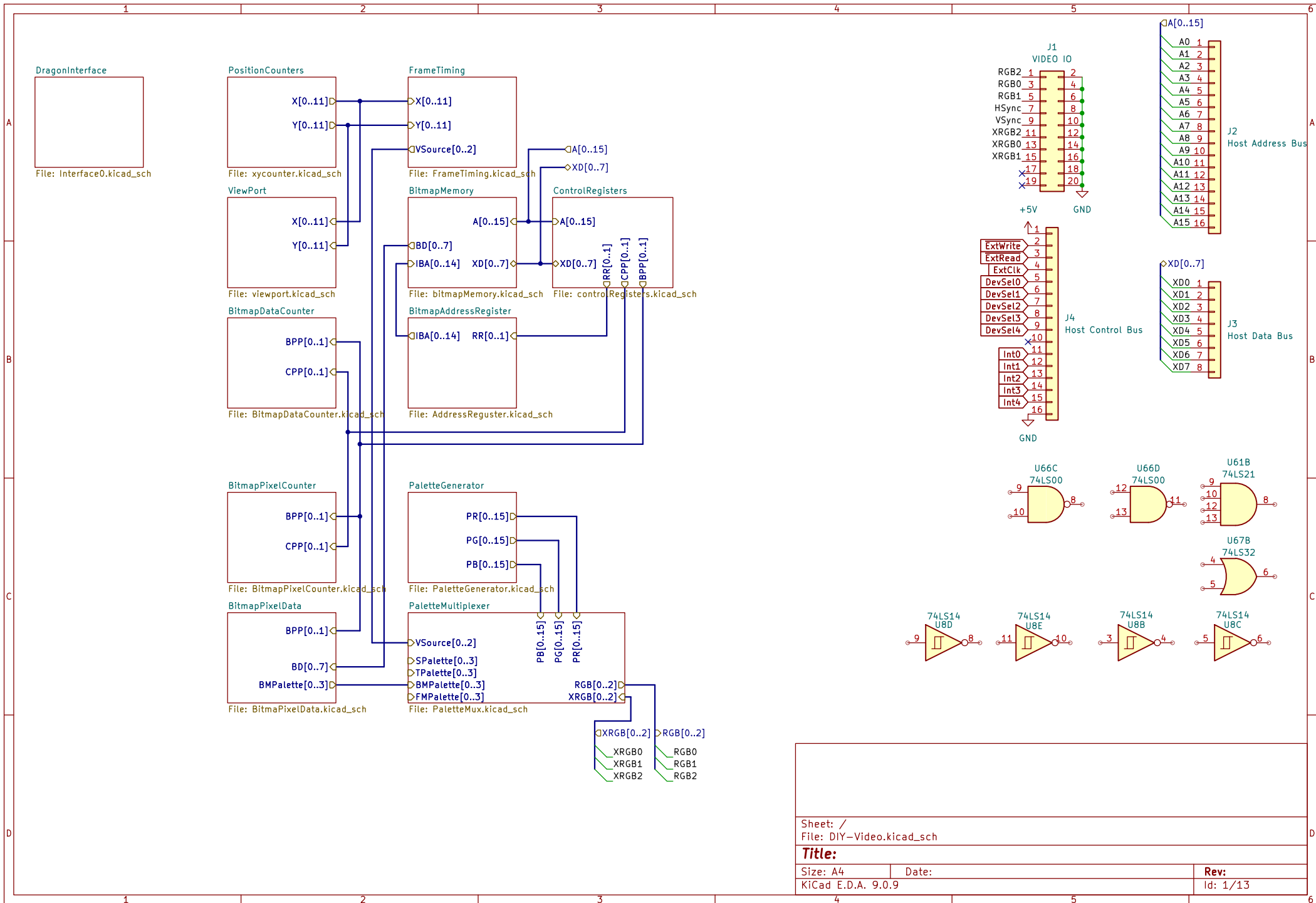
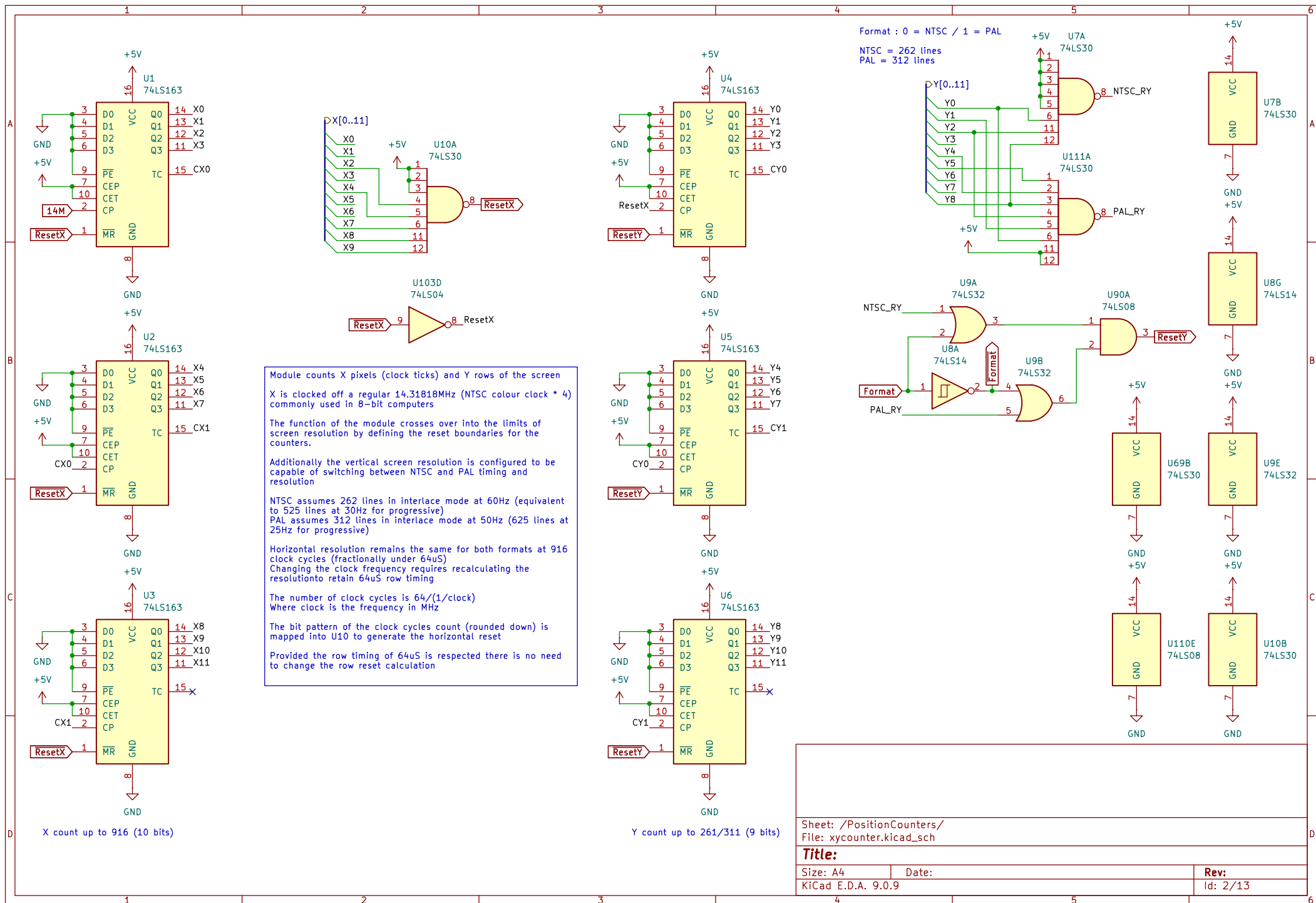
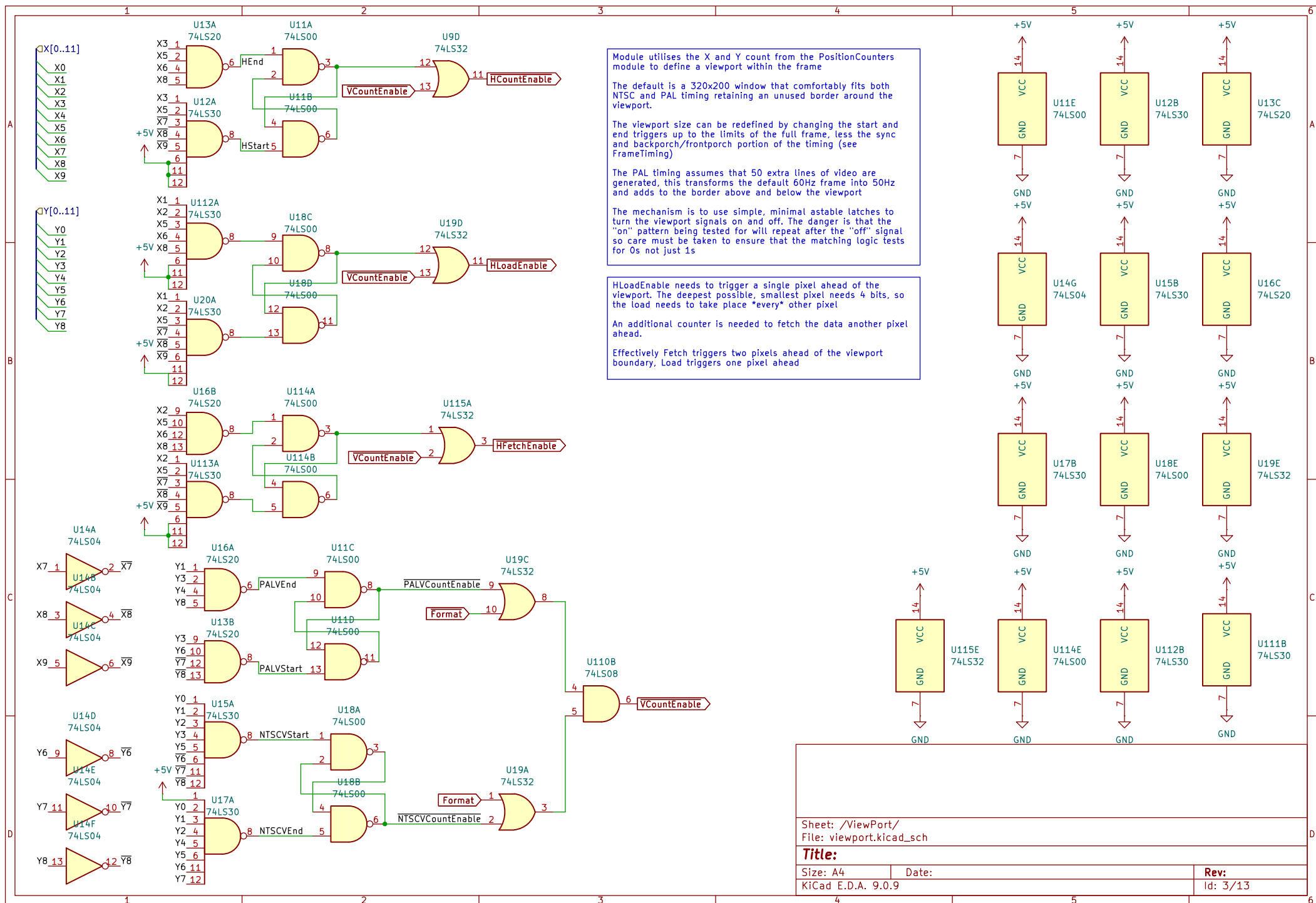
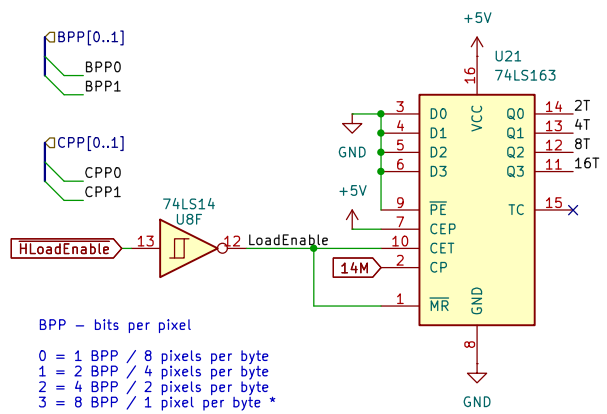










BPP – bits per pixel

0 = 1 BPP / 8 pixels per byte
1 = 2 BPP / 4 pixels per byte
2 = 4 BPP / 2 pixels per byte
3 = 8 BPP / 1 pixel per byte *

* not used

CPP – clocks per pixel

0 = 2 ticks per pixel (HRES = 320)
1 = 4 ticks per pixel (HRES = 160)
2 = 8 ticks per pixel (HRES = 80) *
3 = 16 ticks per pixel (HRES = 40) *

* not used

The BitmapDataCounter module defines when the BitmapPixelData module needs to load a fresh byte of data from the bitmap data bus

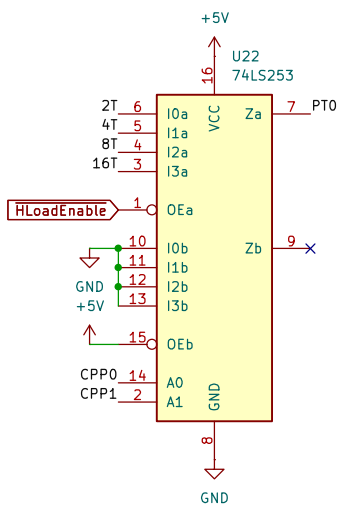
To do calculate when to load the module needs to run a separate counter clocked off the same source as the main pixel counter

The ViewPort module contains a trigger to start the counter out of phase with the data boundary so the data is correctly loaded in advance. Note this does not trigger the memory access, that requires another trigger event in another phase so that the data bus can be prepared before the load action takes place

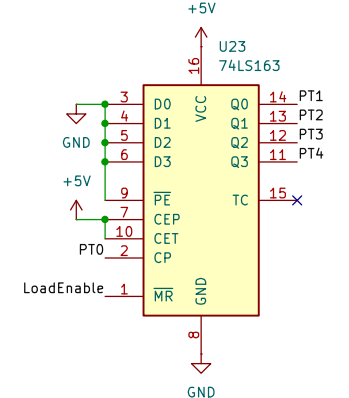
The number of clock ticks per pixel and bits per pixel both impact how frequently the load signal fires. In this simplified design the faster timing options are not used but can be implemented from this design if needed

U21 counts raw clock ticks and acts as a clock divider to reduce the clock by 2, 4, 8 or 16

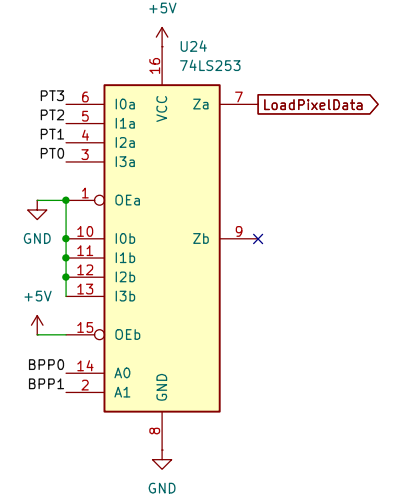
HLoadEnable is a continuous signal, the duration of which is the width of the viewport



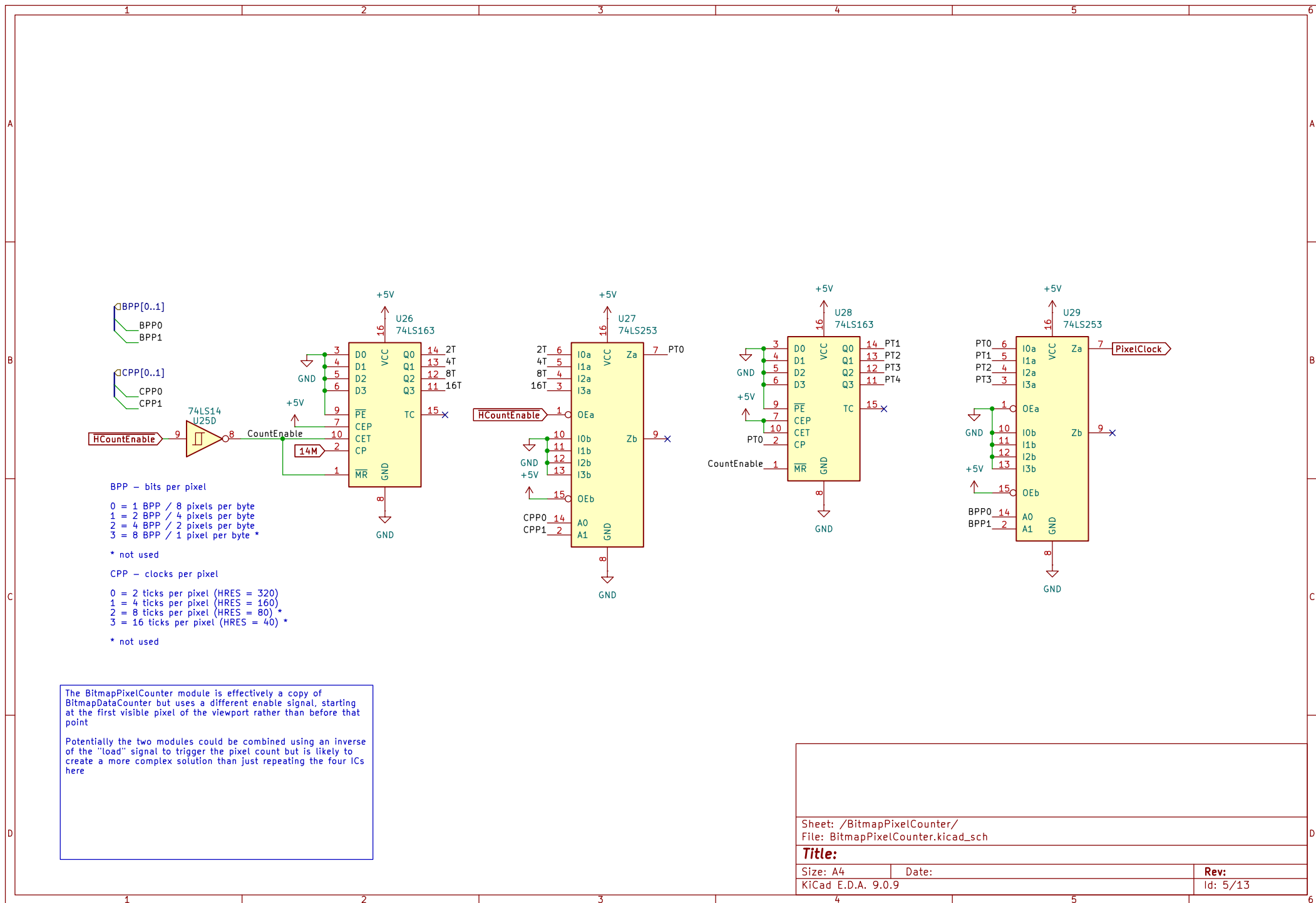
U22 selects the effective clock division based on the clocks per pixel input

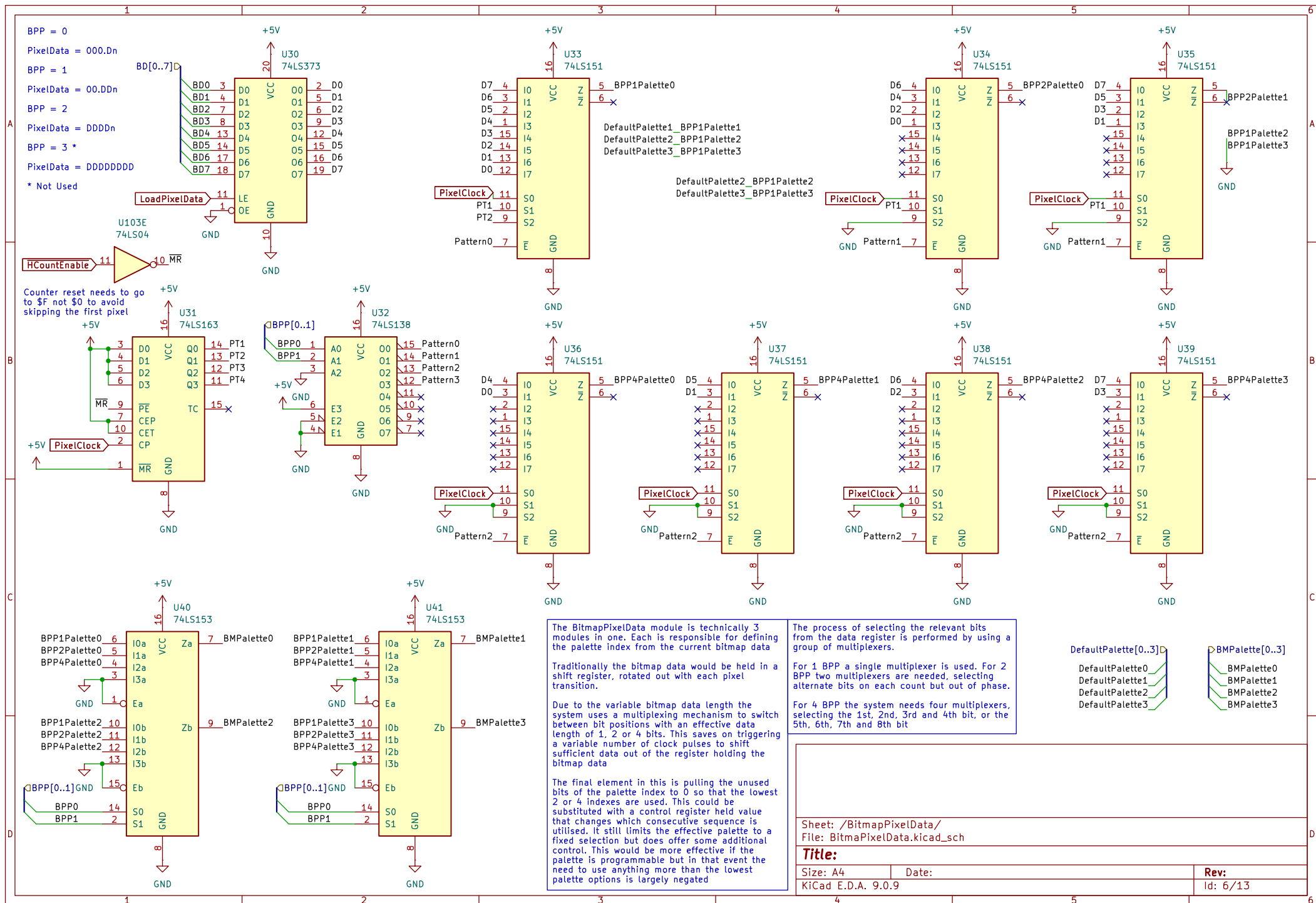


U23 is the effective counter in this circuit, counting pixels



The final stage, U24 selects which pixel count is the effective load pixel data trigger based on the bits per pixel. Note the clock order is reversed as 1 bit per pixel needs more clocks before loading data than 4 bits per pixel





Generate HSync and VSync
Generate Blank

VSyn starts at Y reset
VSyn ends at row 31

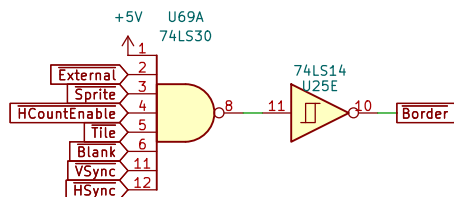
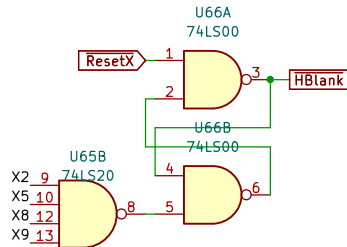
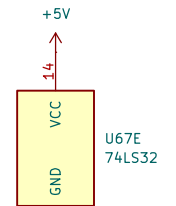
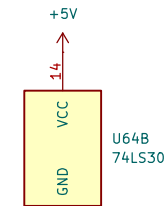
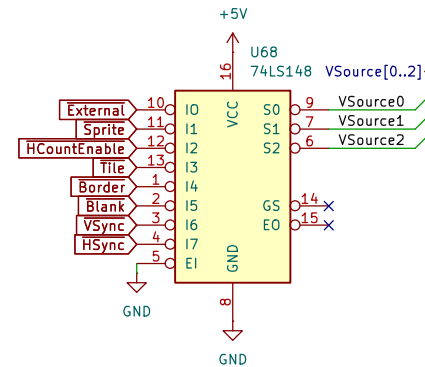
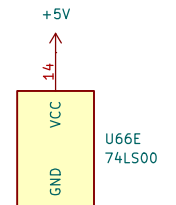
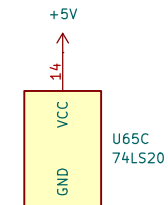
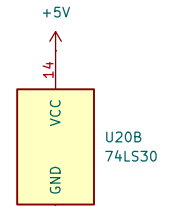
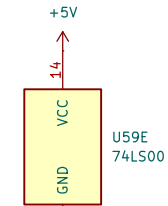
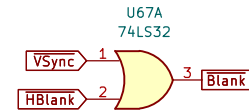
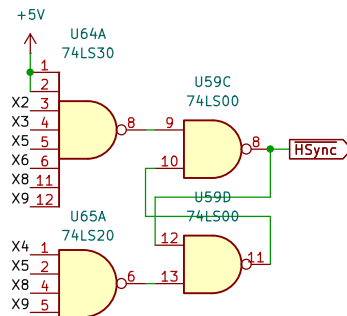
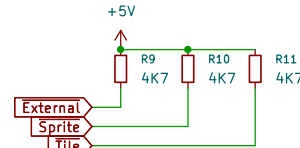
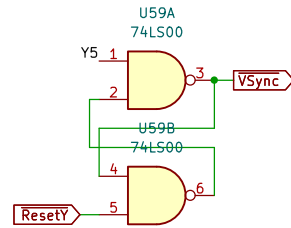
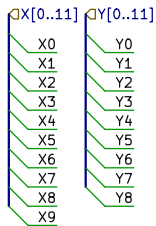
HSync start at column 410
HSync ends at column 437

Blank is VSync or start at Column 402
Blank is VSync or ends at X reset

Generate Output Control Multiplexer

Output Multiplex Control (3 bit)
Using LS148 priority multiplexer

0 = Tile
1 = Bitmap
2 = External
3 = Sprite
4 = Border
5 = Blank
6 = VSync
7 = HSync



Sheet: /FrameTiming/
File: FrameTiming.kicad_sch

Title:

Size: A4

Date:

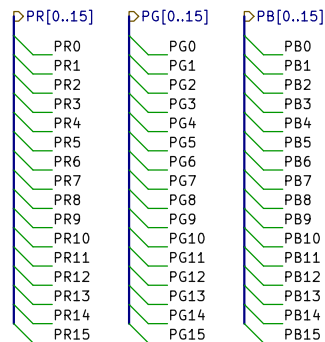
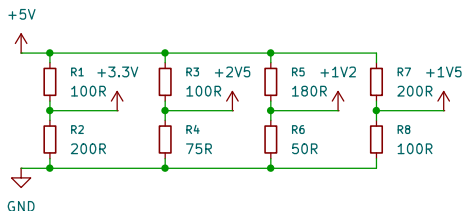
KiCad E.D.A. 9.0.9

Rev:

Id: 7/13

16 Colour Palette

0 = black 0/0/0
1 = white 3.3/3.3/3.3
2 = red 3.3/0/0
3 = blue 0/3.3/0
4 = green 0/0/3.3
5 = yellow = 3.3/3.3/0
6 = magenta = 3.3/0/3.3
7 = cyan = 0/2.2/2.2
8 = d.grey = 1.1/1.1/1.1
9 = l.grey = 2.2/2.2/2.2
10 = m.red = 1.6/0/0
11 = m.blue = 0/0/1.6
12 = m.green = 0/1.6/0
13 = orange = 3.3/1.6/0
14 = violet = 2.2/0/2.2
15 = teal = 0/1.1/1.1



Sheet: /PaletteGenerator/
File: PaletteGenerator.kicad_sch

Title:

Size: A4

Date:

Rev:

KiCad E.D.A. 9.0.9

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